Exploratory Data Analysis - Singapore Total Fertility Rate & Total Live Births

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Introduction

This exploratory data analysis (EDA) investigates the temporal features of two significant population indicators for Singapore:

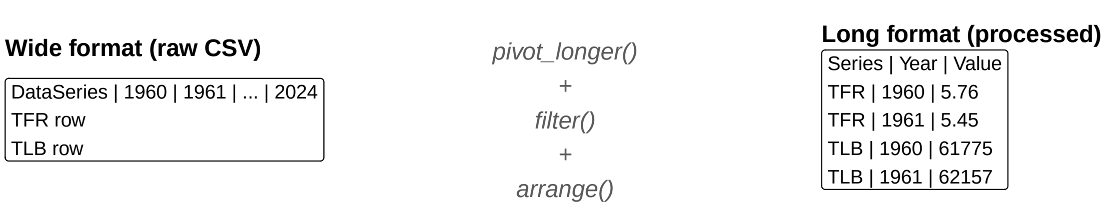
- **Total Fertility Rate (TFR):** the average number of children a woman would have over her lifetime at current age-specific fertility rates;
- **Total Live Births (TLB):** the total number of live births recognised per year.

The dates for the SingaporeCensus.csv reference dataset span from 1960 to 2024, sourced from the Singapore Department of Statistics. The objective is to recognise and analyse long-run trends, assess stationarity, and identify candidate time series models for a proposed training period 1960 - 2012 and a proposed forecasting performance assessed over the remaining 2013 - 2024 period.

Data Loading & Reshaping

The given data is in a CSV file, which is optimal for computation. However, the data is in a 'wide format', where its temporal variables (i.e. years) are given as columns instead of years. We transpose the data and extract the two data columns we are interested in.

Below is the transposition framework of the data.



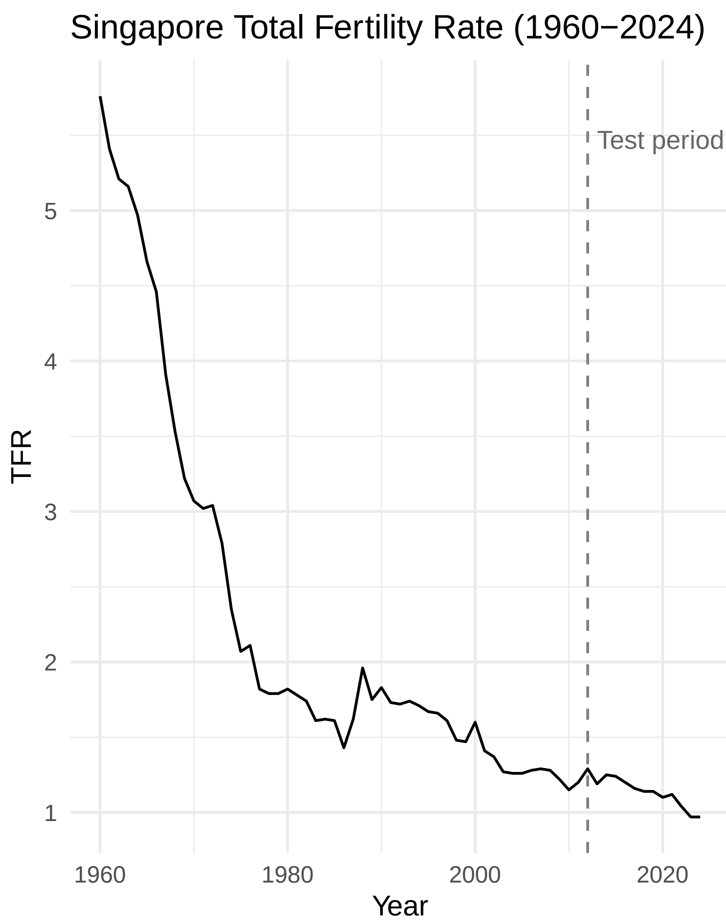
Data Quality & Summary

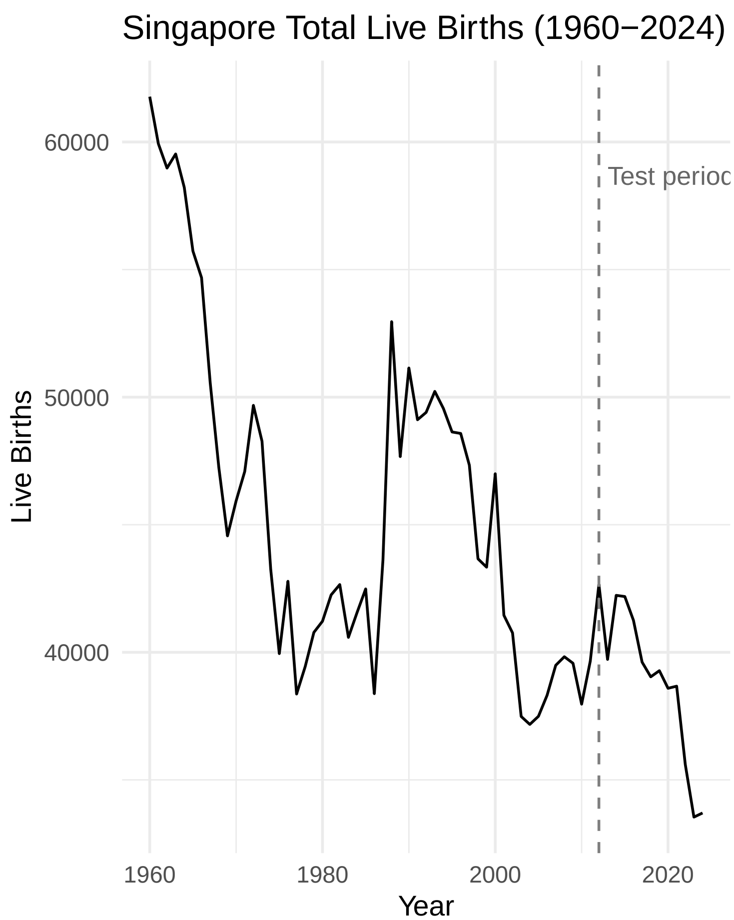
Table 1: Summary statistics for TFR and TLB (1960-2024)

Statistic	TFR	TLB
Min	0.97	33541.00
Q1	1.26	39570.00
Median	1.62	42484.00
Mean	2.06	44359.63
Q3	2.07	48577.00
Max	5.76	61775.00
NAs	0.00	0.00

Time Series Plots

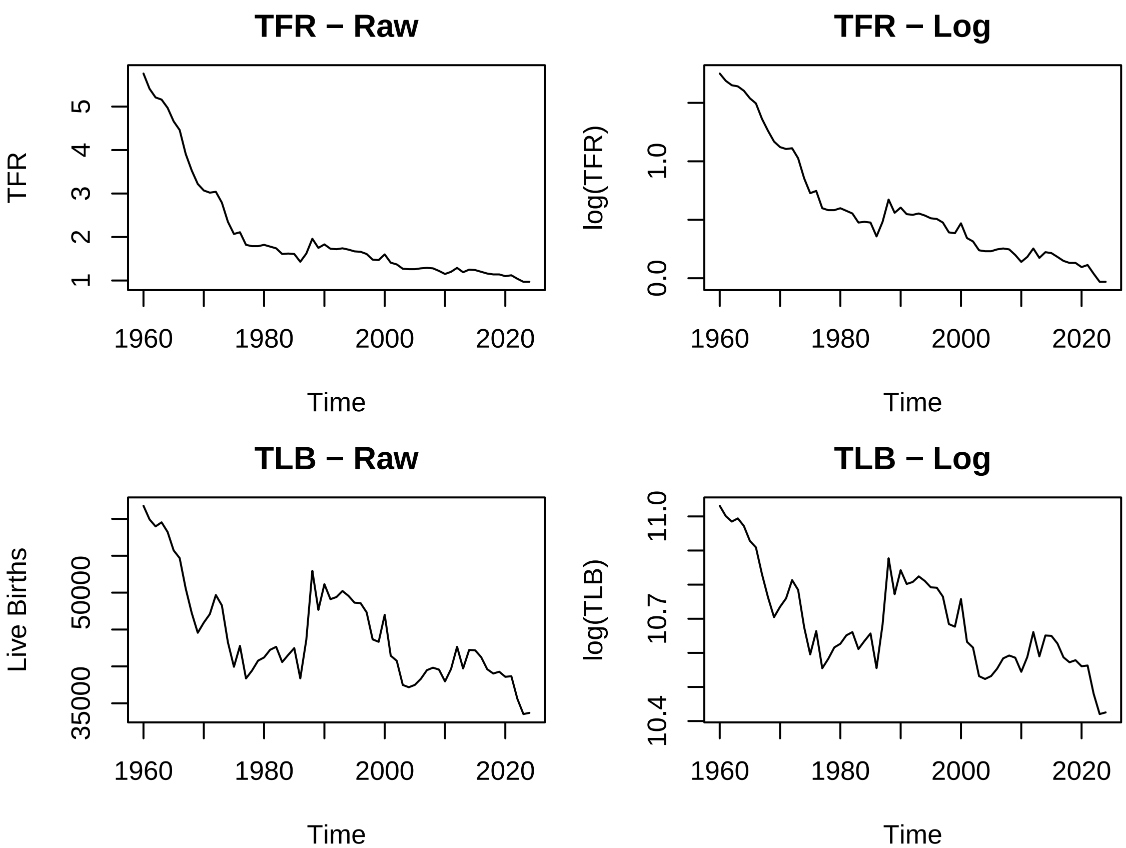
Below, the dashed vertical line marks the train-test split at 2012.




Singapore Total Live Births (1960−2024)
60000
50000
40000
Live Births
Test perioc
1960
1980
2000
2020
Year

Temporal Feature Evaluation

Log Transformation Assessment



Stationarity Testing

Raw Series

Table 2: Stationarity tests - raw training series (1960-2012)

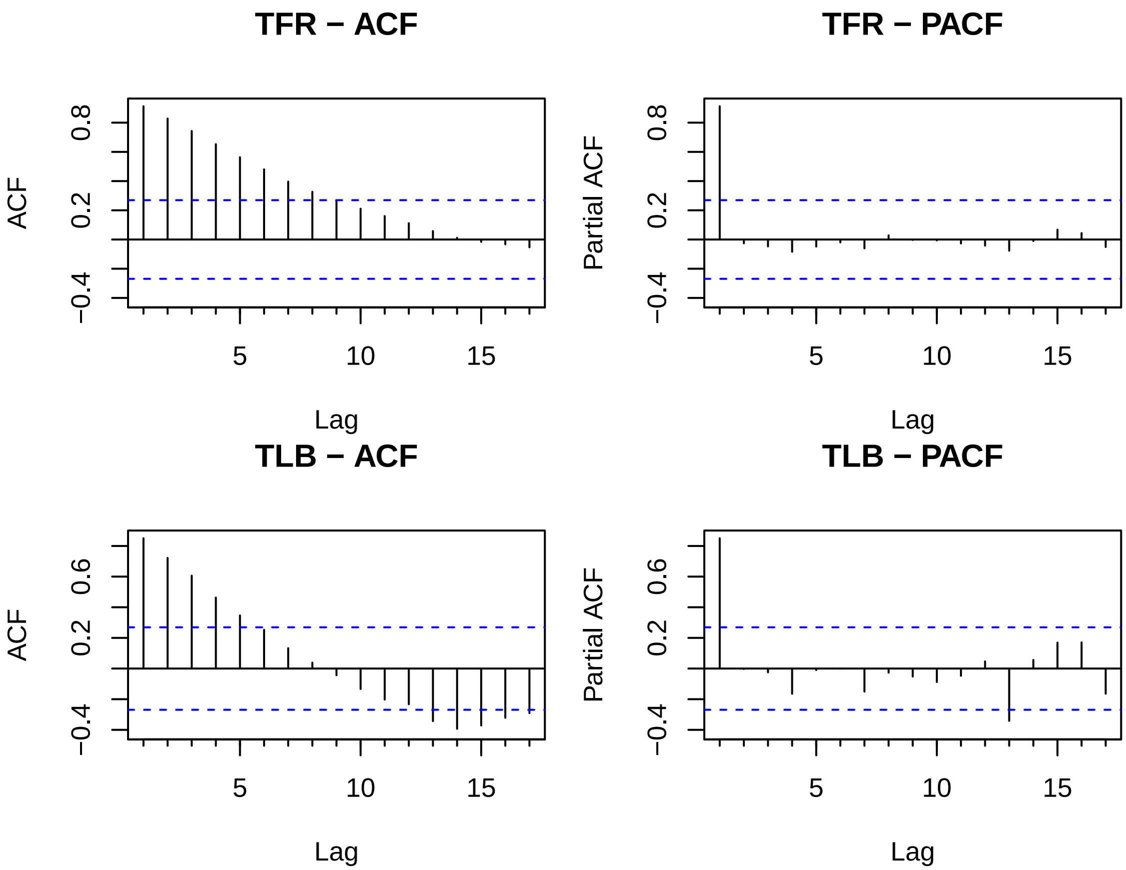
Series	ADF p-value	KPSS p-value
TFR	0.120	0.0100
TLB	0.356	0.0205

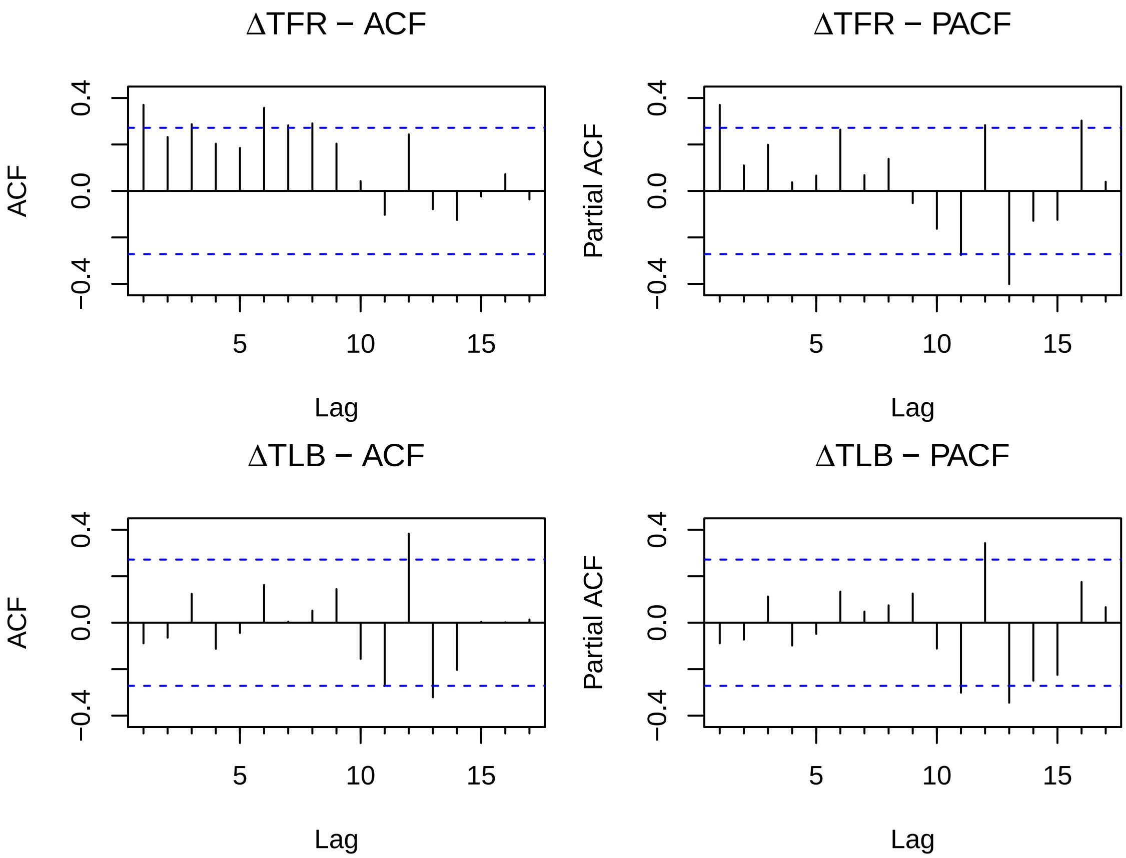
First Difference Series

Table 3: Stationarity tests - first-differenced training series

Series	ADF p-value	KPSS p-value
ΔTFR	0.0830	0.01
ΔTLB	0.0395	0.10

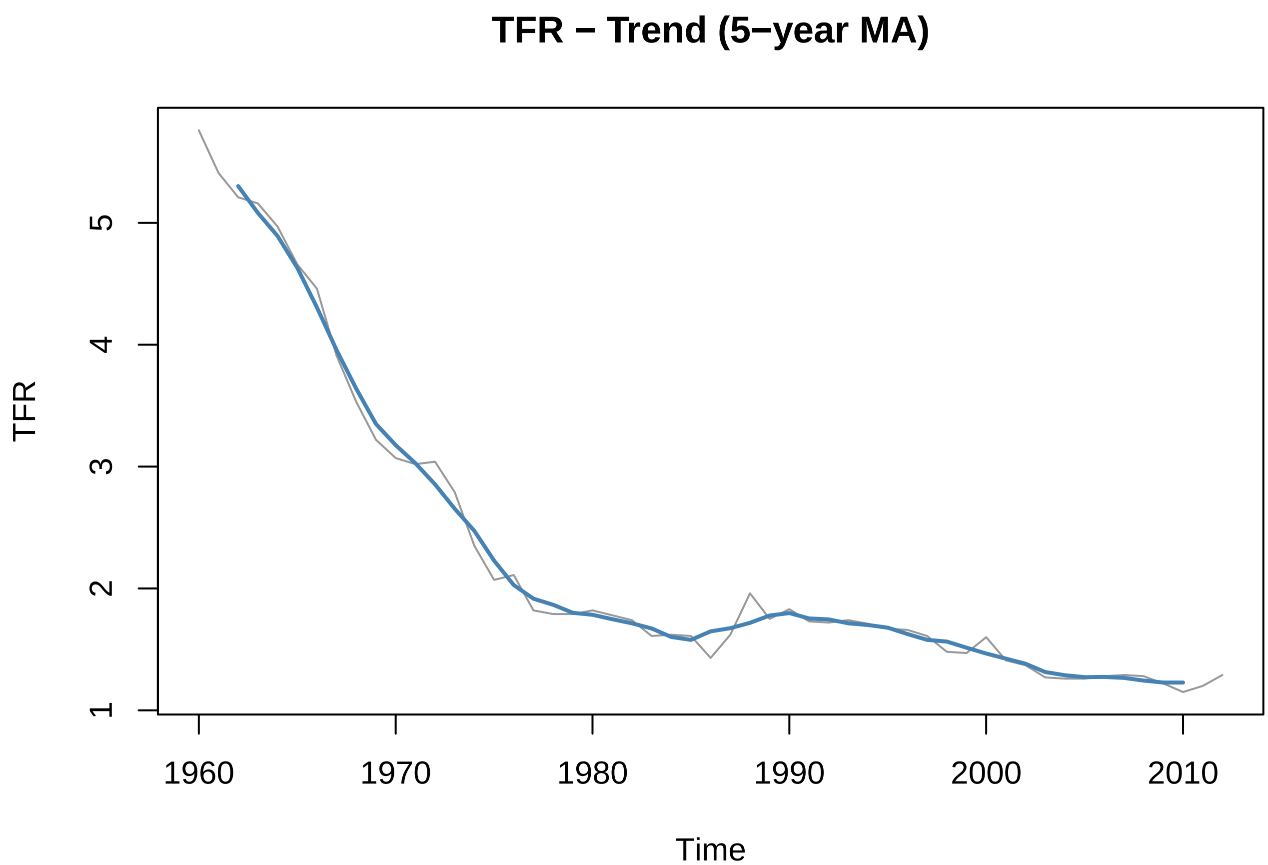
ACF & PACF Analysis

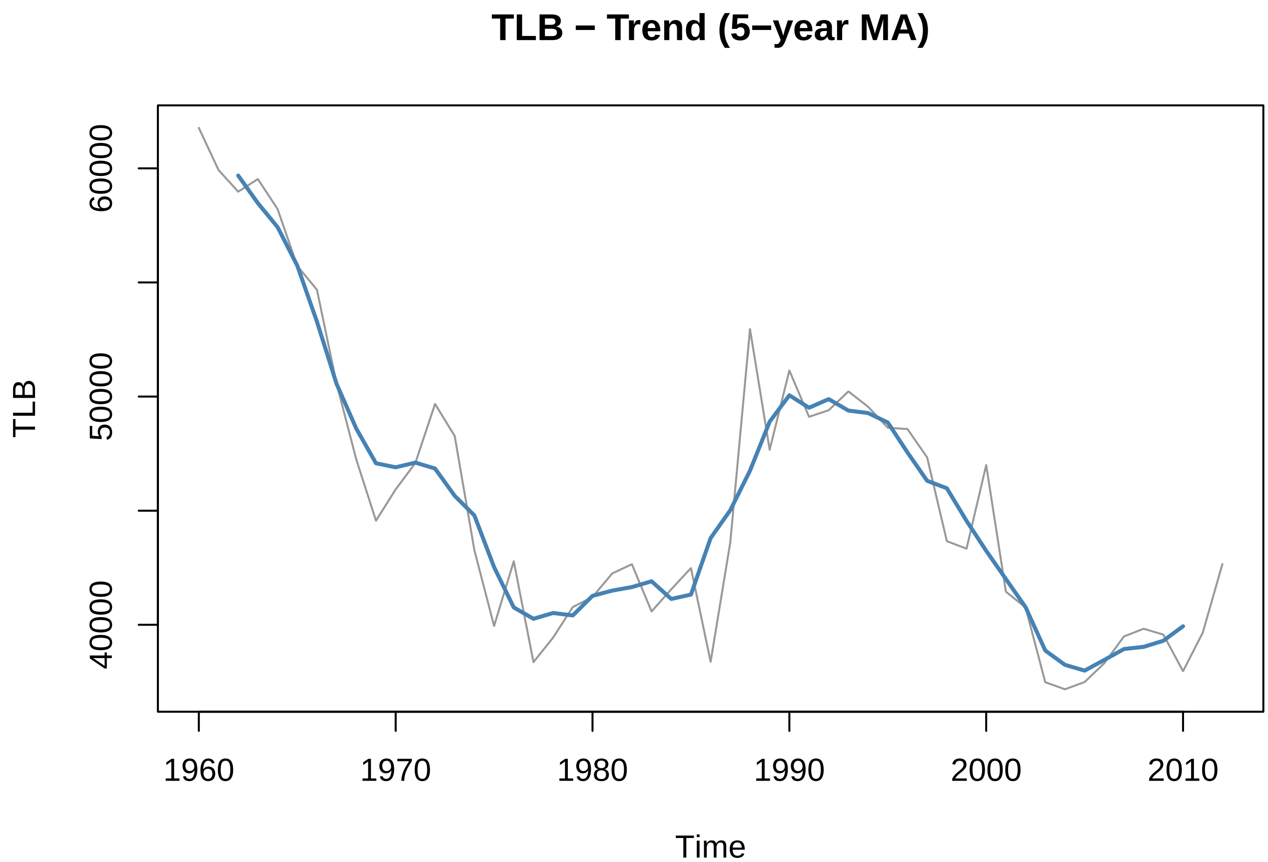




ΔTFR − ACF
ΔTFR − PACF
ΔTLB − ACF
ΔTLB − PACF
ACF
Partial ACF
0.4
0.0
−0.4
5
10
15
Lag

Decomposition





Candidate Model Fitting

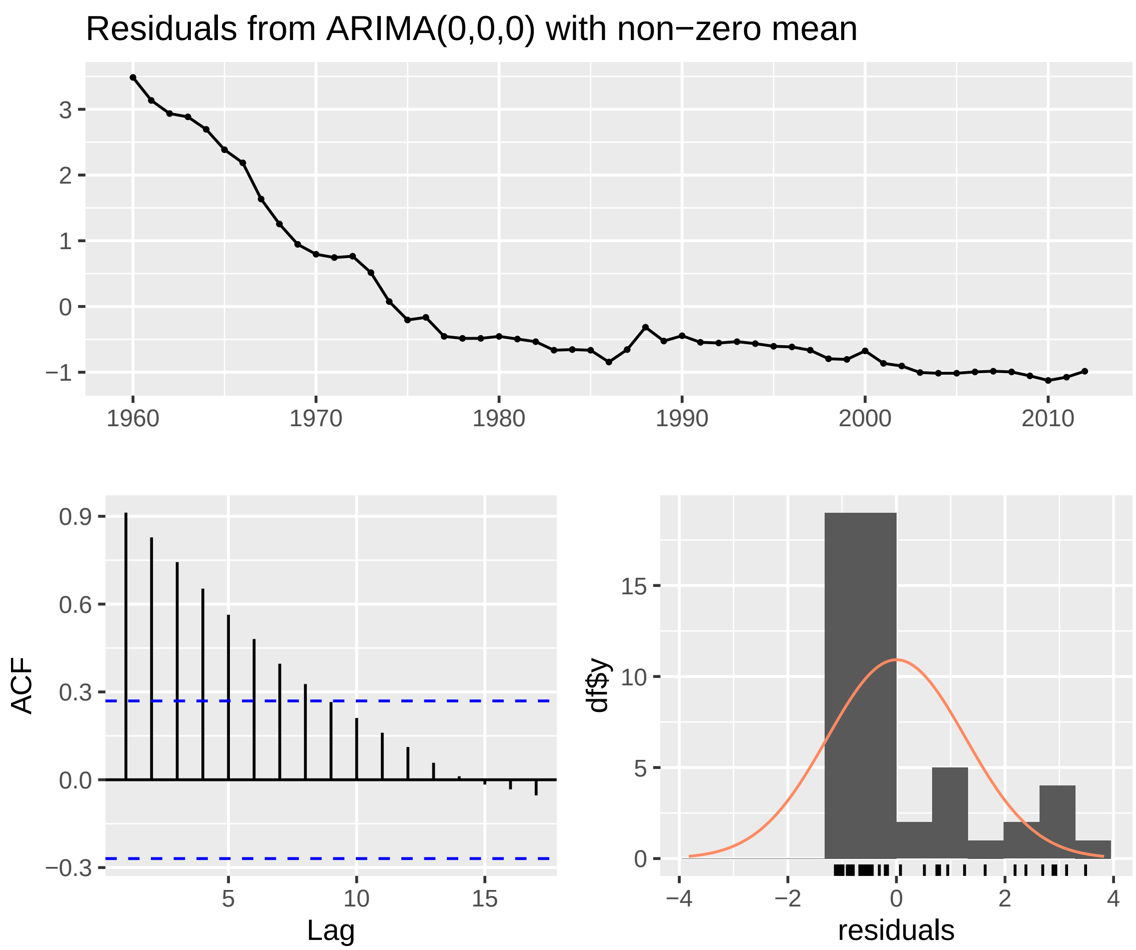
TFR - Candidate Model

Table 4: TFR training set error measures

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0	1.2631	0.9975	-23.1514	45.3835	7.9433	0.9124

Table 5: TFR Ljung-Box test on residuals (lag = 10)

Q*	df	p-value
201.6392	10	0


Residuals from ARIMA(0,0,0) with non−zero mean
ACF
Lag
df$y
residuals

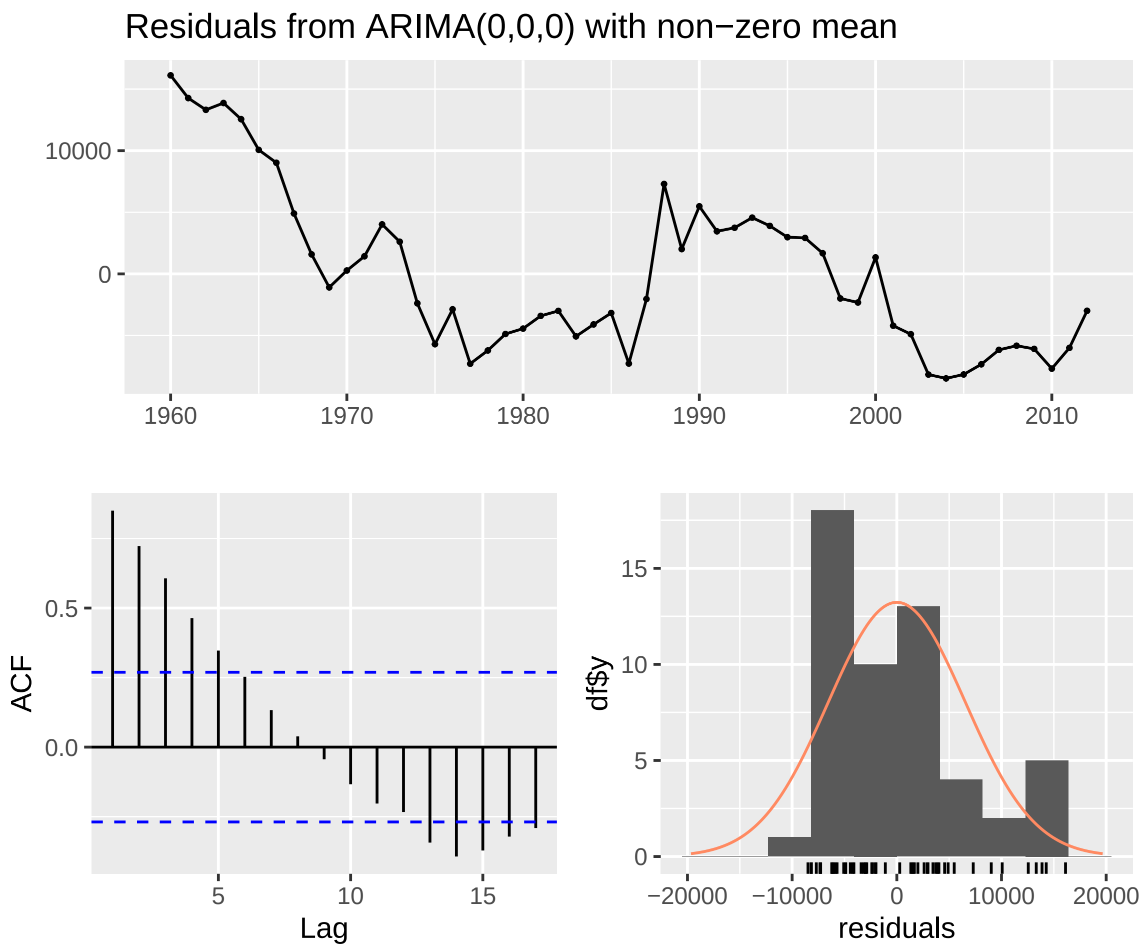
TLB - Candidate Model

Table 6: TLB training set error measures

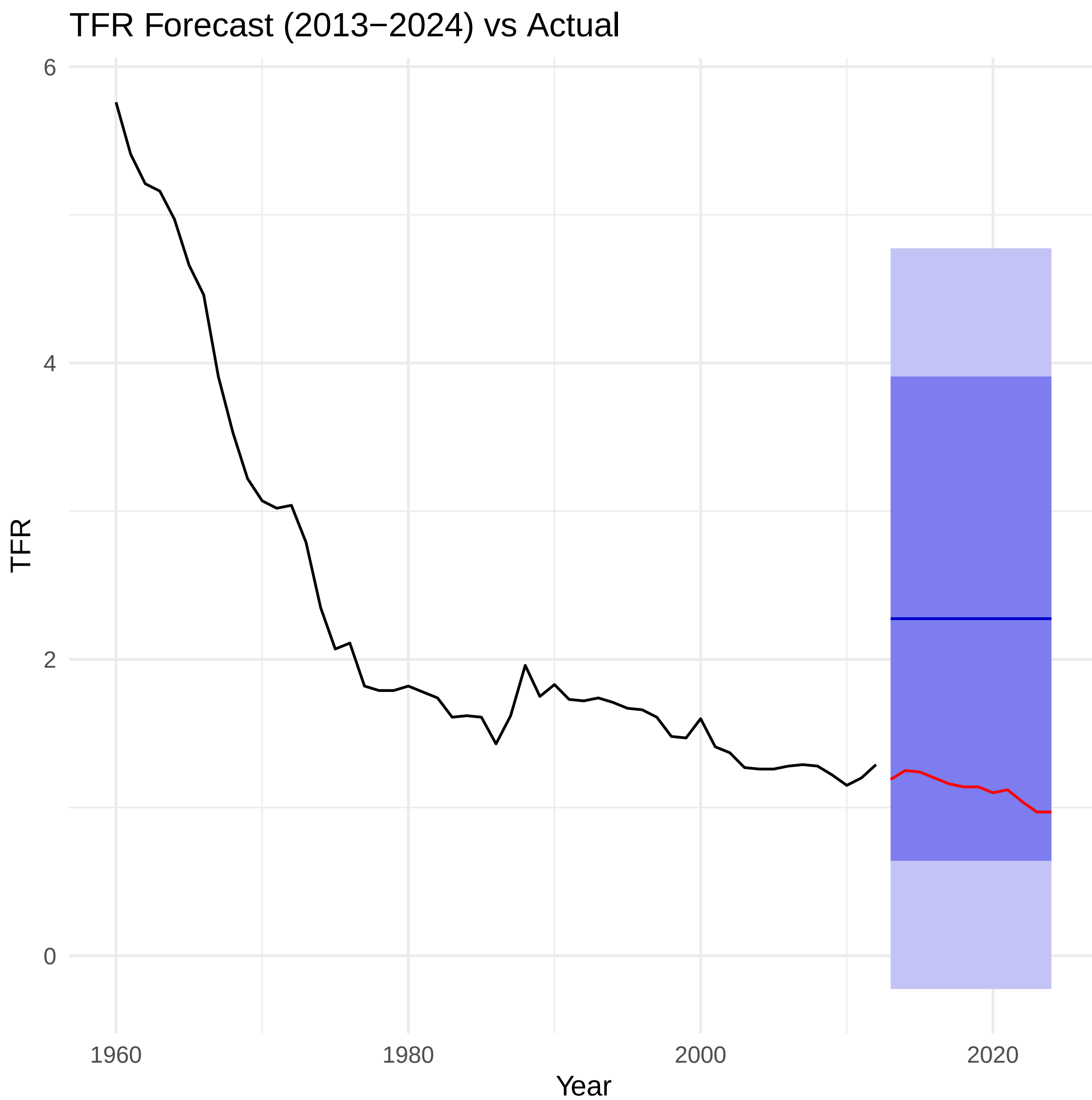
	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0	6492.436	5410.766	-1.888	11.7319	2.5851	0.8504

Table 7: TLB Ljung-Box test on residuals (lag = 10)

Q*	df	p-value
118.416	10	0



Forecasting Model & Assessment



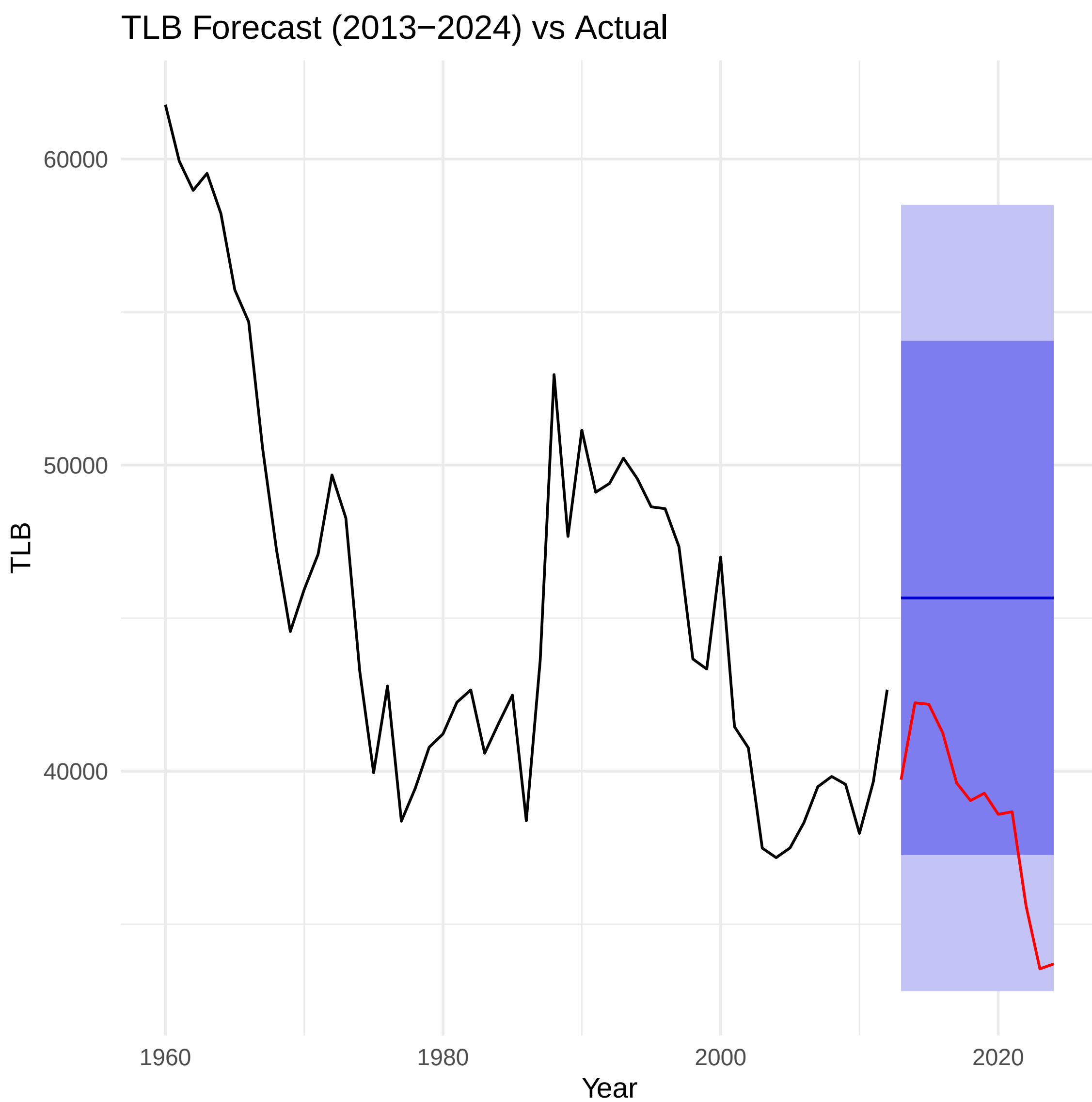


Table 8: TFR forecast accuracy

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	0.0000	1.2631	0.9975	-23.1514	45.3835	7.9433	0.9124	NA
Test set	-1.1484	1.1519	1.1484	-103.2781	103.2781	9.1452	0.7447	26.9001

Table 9: TLB forecast accuracy

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	0.000	6492.436	5410.766	-1.8880	11.7319	2.5851	0.8504	NA
Test set	-7039.987	7578.983	7039.987	-18.8901	18.8901	3.3635	0.7366	5.4094